

Project 1

FortiGate VLAN Security & Python Automation Lab

Prepared by: Mina Alabed

Date: July 2026

Version: 1.0

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1. Project Overview

This project demonstrates the design and implementation of a secure home networking lab using a FortiGate 40F firewall, a managed Layer 2 switch, a Cisco ASR1001-X router, a Dell PC, and a Mac computer. The lab was designed to practice VLAN segmentation, firewall policy creation, secure device management using SSH, and basic Python automation.

The goal of this project is to build a realistic network environment that enforces segmentation, improves security, and automates repetitive tasks. This lab simulates a small enterprise environment where different devices and users are separated into VLANs and controlled by firewall policies.



Purpose

To design, secure, and automate a segmented network using industry-standard equipment and practices.



Lab Environment

- **Internet** → **FortiGate 40F Firewall** → Managed Switch → End Devices
- **VLAN 10:** Dell PC and Mac (User Network)
- **VLAN 20:** Cisco ASR1001-X Router (Lab/Management Network)
- Devices are segmented using VLANs and secured using firewall policies.



Key Features

- | | |
|--|--|
| ✓ VLAN creation and configuration | ✓ Network testing and verification |
| ✓ Inter-VLAN security using FortiGate policies | ✓ Troubleshooting and issue resolution |
| ✓ Web filtering and application control | ✓ Configuration backup and documentation |
| ✓ SSH access for secure device management | ✓ Project documentation and diagrams |
| ✓ Python automation for network tasks | ✓ GitHub ready documentation |



Outcome

By the end of this project, a fully functional and secure segmented network was built, tested, and documented. The project provides hands-on experience with real networking devices, security best practices, and automation using Python scripts.

2. Project Objectives

The main objectives of this project are to design, implement, secure, and automate a small network environment using industry-standard networking equipment and best practices.

These objectives guide the design and configuration of the network and ensure a secure, scalable, and well-documented solution.



1

Build a Segmented Network Using VLANs

Create and configure VLANs to separate different network segments for users and network devices, improving organization and security.



2

Implement Security with FortiGate Firewall

Configure firewall policies to control traffic between VLANs and to the internet, including web filtering and application control.



3

Secure Device Management

Enable and configure SSH access on devices for secure remote management and monitoring.



4

Configure and Manage Switch

Set up the managed switch with VLANs, trunk and access ports, and ensure proper connectivity between all devices.



5

Integrate Cisco ASR1001-X Router

Add the ASR1001-X router to the lab for an additional network segment (VLAN 20) and practice routing and connectivity.



6

Automate Network Tasks with Python

Use Python scripts and APIs (FortiGate REST API) to automate repetitive tasks such as configuration backup and policy verification.



7

Test, Verify, and Troubleshoot

Perform thorough testing to validate connectivity, security policies, and services. Troubleshoot and resolve any issues encountered.



8

Document the Project

Create complete project documentation including topology, configurations, commands, screenshots, and troubleshooting steps for future reference.

3. Lab Equipment



This project was built using physical networking equipment and computers to simulate a small enterprise network. The lab provides hands-on experience with VLAN segmentation, firewall configuration, secure device management, and Python-based network automation.

Hardware

Device	Model	Purpose
	FortiGate 40F	Firewall policies, VLAN interfaces, Internet access, Web Filtering
	Cisco ASR1001-X	Secure device management, SSH access, routing practice
	Managed Layer 2 Switch	VLAN creation, port assignment, network segmentation
	Dell PC	Administration workstation, SSH management, Python automation
	MacBook Air	User workstation, testing, documentation
	Internet Router (Home Router)	Internet connectivity to the lab

Software

Software	Purpose
 FortiGate GUI	Firewall configuration and monitoring
 Web Browser	Access management interfaces
 SSH Client	Secure management of Cisco ASR1001-X
 Python 3	Network automation scripts
 Visual Studio Code	Python script development
 GitHub	Project version control and portfolio
 Apple Pages	Project documentation



Lab Connections

- Internet → Home Router
- Home Router → FortiGate 40F
- FortiGate → Managed Switch (Tagged Port)
- Dell PC → VLAN 10
- MacBook Air → VLAN 20
- Cisco ASR1001-X → VLAN 20



Skills Practiced

- ✓ VLAN Configuration
- ✓ Network Segmentation
- ✓ Firewall Policy Configuration
- ✓ Secure SSH Management
- ✓ Web Filtering
- ✓ Network Documentation
- ✓ Python Automation
- ✓ Troubleshooting
- ✓ GitHub Documentation



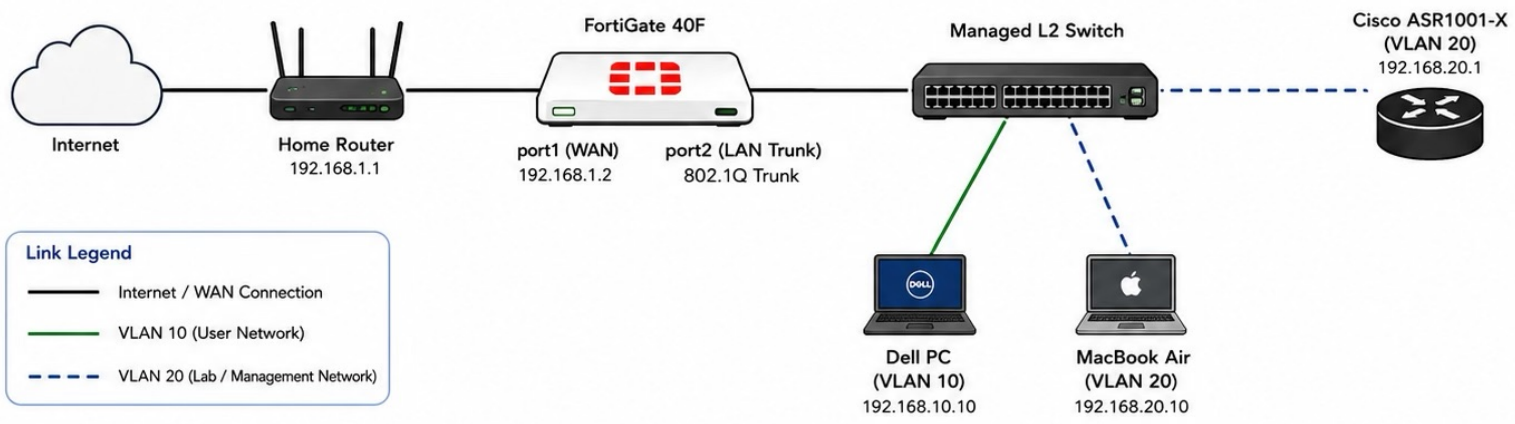
Equipment Summary

This lab combines enterprise networking hardware with automation tools to create a realistic environment for practicing network security concepts. Every configuration was implemented, tested, documented, and prepared for publication on GitHub as a professional portfolio project.



4. Network Topology

The following topology represents the lab environment designed to implement VLAN segmentation, firewall policies, secure device management, and Python automation.





Topology Overview

- Internet connectivity is provided by the Home Router.
- The FortiGate 40F acts as the firewall and default gateway for all VLANs.
- The Managed L2 Switch distributes network segments to end devices using 802.1Q VLANs.
- VLAN 10 is used for user devices (Dell PC).
- VLAN 20 is used for the lab/management network (MacBook Air and Cisco ASR1001-X).



VLAN Summary

VLAN 10 – User Network

- Subnet: 192.168.10.0/24
- Gateway: 192.168.10.1
- Devices: Dell PC

VLAN 20 – Lab / Management Network

- Subnet: 192.168.20.0/24
- Gateway: 192.168.20.1
- Devices: MacBook Air, Cisco ASR1001-X



Key Points

- Inter-VLAN traffic is controlled by FortiGate firewall policies.
- SSH is used to securely manage the Cisco ASR1001-X.
- All configurations are documented and automated using Python scripts.
- This topology simulates a small enterprise network environment.

5. IP Addressing Plan



The IP addressing plan defines all networks, subnets, gateways, and device management addresses used in this lab. Two VLANs are created to separate the user network from the lab/management network.

VLAN 10 – User Network

Purpose	User devices (Dell PC)
VLAN ID	10
Subnet	192.168.10.0/24
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1 (FortiGate VLAN 10 Interface)
DHCP	Disabled (Static IPs used)
Devices	Dell PC

VLAN 20 – Lab / Management Network

Purpose	Lab devices and management (MacBook Air, Cisco ASR1001-X)
VLAN ID	20
Subnet	192.168.20.0/24
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1 (FortiGate VLAN 20 Interface)
DHCP	Disabled (Static IPs used)
Devices	MacBook Air, Cisco ASR1001-X

Addressing Notes

- VLAN 10 is used for all user devices (Dell PC).
- VLAN 20 is used for lab devices and management (MacBook Air & ASR1001-X).
- FortiGate provides routing between VLANs and access to the internet.
- All IP addresses are within private address ranges (RFC1918).

Device IP Addressing

Device	Interface / Port	IP Address	Subnet Mask	VLAN	Purpose
 FortiGate 40F	port1 (WAN)	192.168.1.2	255.255.255.0	N/A	Internet (WAN)
	port2 (LAN Trunk)	N/A	N/A	10, 20	Trunk to Switch
	VLAN 10 Interface	192.168.10.1	255.255.255.0	10	Gateway for User Network
	VLAN 20 Interface	192.168.20.1	255.255.255.0	20	Gateway for Lab Network
 Managed L2 Switch	Management (VLAN 1)	192.168.1.10	255.255.255.0	1	Switch Management
 Dell PC	NIC	192.168.10.10	255.255.255.0	10	User Device
 MacBook Air	NIC	192.168.20.10	255.255.255.0	20	User Device
 Cisco ASR1001-X	Gig0/0/0	192.168.20.100	255.255.255.0	20	Lab Router

Subnet Summary

- VLAN 10: 192.168.10.0/24 → Gateway: 192.168.10.1
- VLAN 20: 192.168.20.0/24 → Gateway: 192.168.20.1
- WAN Network: 192.168.1.0/24 (Provided by Home Router)

6. VLAN Configuration



VLANs are used to segment the network into separate broadcast domains for security, organization, and traffic management. The managed switch uses 802.1Q trunks to carry multiple VLANs between devices.

VLAN Overview

- **VLAN 10** – User Network (Dell PC)
- **VLAN 20** – Lab / Management Network (MacBook Air & Cisco ASR1001-X)
- **VLAN 1** – Default Management VLAN

VLAN Information

VLAN ID	VLAN Name	Purpose	Subnet / Gateway	Interfaces
10	USER	User devices (Dell PC)	192.168.10.0/24 GW: 192.168.10.1	Untagged: Port 3 Tagged: Port 1 (Trunk)
20	LAB_MGMT	Lab devices & management (MacBook Air, ASR1001-X)	192.168.20.0/24 GW: 192.168.20.1	Untagged: Port 5, Port 6 Tagged: Port 1 (Trunk)
1	DEFAULT	Switch management (Default VLAN)	192.168.10.200/24	Untagged: Port 1 (Management Access)

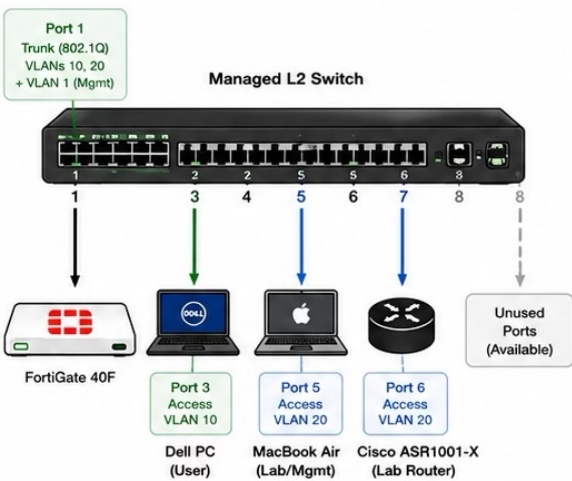
Switch Port Configuration

Port	Connected Device	Mode	Access VLAN (Untagged)	Trunk VLANs (Tagged)	Description
1	FortiGate 40F	Trunk	VLAN 1 (Mgmt)	10, 20	802.1Q Trunk to Firewall (also switch management)
2	(Unused)	Access	VLAN 1	-	Available
3	Dell PC	Access	VLAN 10	-	User Network
4	(Unused)	Access	VLAN 1	-	Available
5	MacBook Air	Access	VLAN 20	-	Lab / Management Network
6	Cisco ASR1001-X	Access	VLAN 20	-	Lab / Management Network
7-8	(Unused)	Access	VLAN 1	-	Available

Key Points

- Port 1 is a trunk port carrying VLAN 10 and VLAN 20 to the FortiGate.
- Ports 3 (Dell PC) is in VLAN 10 for user traffic.
- Ports 5 (MacBook Air) and 6 (ASR1001-X) are in VLAN 20 for lab/management traffic.
- VLAN 1 is used only for switch management.

Port Role Diagram



Benefits of this VLAN Design

- ✓ Improved security by isolating user and lab networks.
- ✓ Better traffic control and reduced broadcast domains.
- ✓ Easier management and scalability for future expansion.

7. Firewall Policies (FortiGate 40F)



Firewall policies are configured on the FortiGate 40F to control traffic between VLANs and the internet. The goal is to allow necessary services, block unwanted access, and protect the internal network.

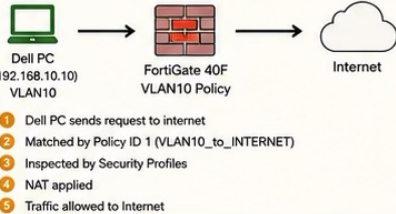
Firewall Policy Table

ID	Name	Source	Destination	Service	Action	Schedule	NAT	Security Profiles	Log	Purpose
1	VLAN10_to_INTERNET	VLAN10 (192.168.10.0/24)	WAN	ALL	ACCEPT	always	Enabled	UTM, AV, Web Filter	All	Allow VLAN 10 users to access the internet
2	VLAN20_to_INTERNET	VLAN20 (192.168.20.0/24)	WAN	ALL	ACCEPT	always	Enabled	UTM, AV, Web Filter	All	Allow VLAN 20 lab devices to access the internet
3	VLAN10_to_VLAN20	VLAN10 (192.168.10.0/24)	VLAN20 (192.168.20.0/24)	ALL	ACCEPT	always	Disabled	None	All	Allow user network access to lab network
4	VLAN20_to_VLAN10	VLAN20 (192.168.20.0/24)	VLAN10 (192.168.10.0/24)	ALL	DENY	always	Disabled	None	All	Block lab network from accessing user network
5	SSH_to_ASR	VLAN10 (192.168.10.0/24)	ASR1001-X (192.168.20.100)	SSH (TCP 22)	ACCEPT	always	Disabled	None	All	Allow SSH access to ASR from user network
6	WEB_FILTER_Block_FB	VLAN10, VLAN20	WAN	HTTPS (TCP 443)	ACCEPT	always	Enabled	Web Filter (Block FB)	All	Web filtering rule to block Facebook
7	IMPLICIT_DENY	all	all	ALL	DENY	always	Disabled	None	All	Deny all other traffic by default

Policy Flow Summary

- 1 Traffic from internal VLANs to the internet is allowed and inspected.
- 2 VLAN10 can access VLAN20 (lab) network.
- 3 VLAN20 is blocked from accessing VLAN10 for security.
- 4 SSH access to ASR is allowed only from VLAN10.
- 5 Web filtering blocks Facebook for both VLANs.
- 6 All other traffic is denied by the implicit deny rule.

Example: VLAN10 to Internet



Policy Overview

- ✓ Policies are evaluated top-down (ID order).
- ✓ Implicit deny rule at the end blocks all others.
- ✓ Logging is enabled for monitoring and troubleshooting.
- ✓ NAT is enabled for internet access (outgoing traffic).

Network Zones / Interfaces

Interface	Name / Zone	IP Address	Role
port1	WAN	192.168.1.2/24	Internet (from Home Router)
port2	LAN (Trunk)	N/A (802.1Q)	Trunk to Switch (VLANs 10, 20, 1)
VLAN10	VLAN10	192.168.10.1/24	Gateway for User Network
VLAN20	VLAN20	192.168.20.1/24	Gateway for Lab Network

Security Profiles Used

- UTM Profile: IPS, Application Control, Antivirus, Web Filter
- Web Filter Profile: Facebook (facebook.com) blocked
- Logging: All allowed and blocked traffic is logged
- NAT: Enabled for internet access (policy based)

Notes

- Implicit deny ensures any traffic not explicitly allowed is blocked.
- All policies are logged for visibility and troubleshooting.
- You can add more granular policies as the network grows (e.g., DNS, NTP, specific application rules).
- Policy order is important. Always place specific allow policies above broader ones.

8. Testing & Verification



All configurations were tested to verify connectivity, VLAN segmentation, firewall policies, and secure access to devices. The results confirm that the network is working as designed.

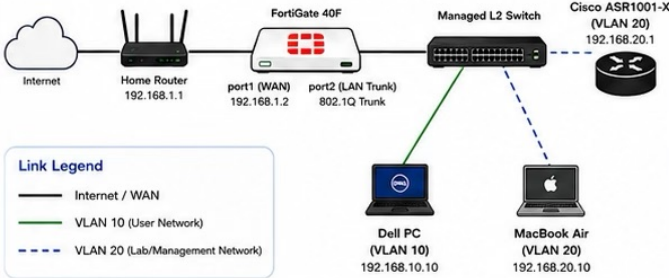
Connectivity Test Matrix

Test	Source	Destination	Protocol / Service	Expected Result	Actual Result	Status
1	Dell PC (VLAN 10)	MacBook Air (VLAN 20)	Ping (ICMP)	Blocked	No Reply	✓
2	MacBook Air (VLAN 20)	Dell PC (VLAN 10)	Ping (ICMP)	Blocked	No Reply	✓
3	Dell PC (VLAN 10)	Internet	HTTP/HTTPS	Allowed	Success	✓
4	MacBook Air (VLAN 20)	Internet	HTTP/HTTPS	Allowed	Success	✓
5	Dell PC (VLAN 10)	Cisco ASR1001-X (VLAN 20)	SSH (TCP 22)	Allowed	Success	✓
6	Web Access (VLAN 10)	facebook.com	HTTPS (TCP 443)	Blocked	Blocked	✓
7	Web Access (VLAN 20)	facebook.com	HTTPS (TCP 443)	Blocked	Blocked	✓

Test Summary

- ✓ VLAN configuration: Passed
 - ✓ Inter-VLAN traffic control: Passed
 - ✓ Internet access (VLAN 10): Passed
- ✓ SSH access to ASR1001-X: Passed
 - ✓ Web Filtering (Facebook blocked): Passed

Test Topology Reference



Test Results Screenshots

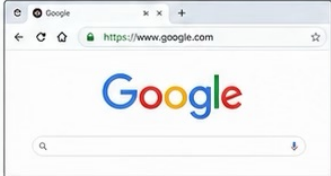
1. Dell PC → MacBook Air (Ping Blocked)

```
C:\> ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Result: Blocked ❌

2. Internet Access (Dell PC) (Success)



Result: Success ✓

3. SSH to ASR1001-X (Success)

```
C:\> ssh -l admin 192.168.20.1
admin@192.168.20.1's password:
Last login: Tue Jul 1 21:30:10 2026
ASR1001-X#
ASR1001-X# show ip interface brief
Interface      IP-Address    OK? Method Status  Protocol
GigabitEthernet0/0/0  192.168.20.1 YES manual up      up

ASR1001-X#
```

Result: Success ✓

4. Web Filtering (Facebook Blocked) (Blocked)



Result: Blocked ❌

What We Verified

- ✓ VLANs are working and devices are isolated by design.
- ✓ Firewall policies are correctly controlling inter-VLAN traffic.
- ✓ Internet access is available for both VLANs.
- ✓ SSH access to ASR1001-X is allowed only from VLAN 10.
- ✓ Web filtering is applied and Facebook is blocked.

Notes

- All tests were performed after saving the configuration on FortiGate and the Switch.
- If any test fails in the future, review VLAN assignments, firewall policies, and interface status.
- Logs on FortiGate were checked to confirm policy hits for allowed and blocked traffic.
- Screenshots and logs are saved for documentation and troubleshooting.

9. Python Automation



Python automation is used to simplify device management, backup configurations, and monitor the network. The script connects to the FortiGate 40F via its REST API to retrieve system information and save the running configuration automatically.



Automation Goals

- ✓ Connect to FortiGate 40F using REST API.
- ✓ Backup the running configuration.
- ✓ Retrieve system information (hostname, version, interfaces, policies).
- ✓ Save output to local files for documentation.
- ✓ Build a reusable automation framework for future tasks.

Python Script Overview

```
import requests
import json

FGT_IP = "192.168.10.1"
USER = "admin"
PASSWORD = "YourStrongPassword"

# Disable SSL warning
requests.packages.urllib3.disable_warnings()

# Login to FortiGate
login_url = f"https://{FGT_IP}/api/v2/logincheck"
payload = {"username": USER, "secretkey": PASSWORD}
headers = {"Content-Type": "application/json"}

response = requests.post(login_url, json=payload, headers=headers, verify=False)
response.raise_for_status()
token = response.json()['session']
headers["Authorization"] = f"Bearer {token}"

# Get system information
sys_url = f"https://{FGT_IP}/api/v2/monitor/system/status"
sys_info = requests.get(sys_url, headers=headers, verify=False).json()

# Get configuration backup
backup_url = f"https://{FGT_IP}/api/v2/monitor/system/config/backup"
backup = requests.get(backup_url, headers=headers, verify=False).json()

# Save outputs
with open("fortigate_system_info.json", "w") as f:
    json.dump(sys_info, f, indent=4)

with open("fortigate_backup.txt", "w") as f:
    f.write(backup['data'])

print("✅ Automation completed successfully!")
```



What the Script Does

- 1 Authenticates to the FortiGate using API credentials.
- 2 Retrieves system information (hostname, version, uptime, etc.).
- 3 Downloads the running configuration backup.
- 4 Saves the outputs to local files.
- 5 Provides a foundation for future automation tasks (policy check, interface status, logs, etc.).

Sample Output

1. System Information (fortigate_system_info.json)

```
{
  "hostname": "FortiGate-40F",
  "serial-no": "FGT40FTK22012345",
  "version": "v7.2.6",
  "mode": "nat",
  "system-time": "2026-07-01 20:30:45",
  "uptime": 86400,
  "cpu-usage": 12,
  "memory-usage": 35
}
```

2. Configuration Backup (fortigate_backup.txt)

```
config system global
  set hostname "FortiGate-40F"
  set timezone 04
  set admin-sport 443
end

config system interface
  edit "VLAN10"
    set ip 192.168.10.1 255.255.255.0
  next
  edit "VLAN20"
    set ip 192.168.20.1 255.255.255.0
  next
end
...
(config continues)
```

How to Run

- 1 Make sure Python 3 is installed on your system.
- 2 Install required library:
pip install requests
- 3 Update the script with your FortiGate IP and admin password.
- 4 Run the script:
python fortigate_backup.py
- 5 Check the generated files in the same folder.

Generated Files

- fortigate_system_info.json
→ System information in JSON format
- fortigate_backup.txt
→ Running configuration backup

These files are used for documentation, troubleshooting, and version tracking.



Benefits

- ✓ Saves time by automating repetitive tasks.
- ✓ Reduces human errors in manual backups.
- ✓ Provides up-to-date configuration backup.
- ✓ Can be scheduled to run automatically (cron job / Task Scheduler).
- ✓ Foundation for advanced automation (policy audit, interface monitoring, alerts).



Next Automation Ideas

- ✓ Check interface status and log link down events.
- ✓ Retrieve and analyze firewall logs.
- ✓ Verify policy hit counts.
- ✓ Automate report generation and email delivery.
- ✓ Push configuration changes (future phase).



Notes

- Always secure your API credentials.
- Use read-only admin account if possible.
- Store passwords securely or use environment variables.
- Test scripts in a safe lab environment before production.

10. Troubleshooting



During the implementation of the lab, several issues were encountered. This section documents the problems, root causes, and the steps taken to resolve them.



Troubleshooting Approach

- Verify physical connections first.
- Check IP addressing and VLAN assignments.
- Review device configurations and firewall policies.
- Use logs and test tools to identify the root cause.
- Document the issue and resolution for future reference.

Common Issues and Resolutions

#	Issue	Symptoms	Root Cause	Resolution	Status
1	Cannot access FortiGate GUI (192.168.10.1)	GUI page not loading on Chrome/Safari	Browser blocking mixed content / certificate or cached data	- Used Firefox browser - Cleared cache and cookies - Accessed via https://192.168.10.1	Resolved
2	No internet access from VLAN 10	- Cannot browse internet - DNS not resolving	Missing firewall policy for VLAN 10 to WAN	- Created policy: VLAN10 → WAN (ALL) - Enabled NAT - Allowed DNS (port 53)	Resolved
3	Cannot ping between VLAN 10 and VLAN 20	Ping fails both ways	No inter-VLAN routing or policy to allow traffic	- Created policies to allow VLAN10 → VLAN20 - Blocked VLAN20 → VLAN10 as per goal - Verified policies order	Resolved
4	Web filter not working (Facebook still open)	Facebook page loads even though it's blocked	QUIC protocol (UDP/443) bypasses web filter	- Created firewall policy to block QUIC (UDP/443) - Forced TCP for web traffic - Verified in logs	Resolved
5	SSH to ASR1001-X failed from Dell PC	Connection timeout or refused	SSH version / encryption mismatch	- Enabled SSH v2 on ASR - Configured compatible ciphers and KEX - SSH connection succeeded	Resolved
6	Lost switch management access	Cannot access 192.168.10.200	VLAN 1 removed from management port	- Re-added VLAN 1 (untagged) to Port 1 - Restored management access	Resolved

FortiGate Log Example (Web Filter Blocked)

date/time	action	proto	src ip	dst ip	service	app	policyid	msg
2026-07-01 14:22:10	deny	TCP	192.168.10.10	157.240.22.35	HTTPS	facebook	6	webfilter
2026-07-01 14:22:11	deny	QUIC	192.168.10.10	157.240.22.35	QUIC	facebook	6	quic-block

(Logs confirm Facebook access is blocked)

Switch Port Status (Example)

Port Status				
Port	Link	Mode	Access VLAN (Untagged)	Trunk VLANs (Tagged)
1 (FortiGate)	↑	Trunk	VLAN 1	10, 20
3 (Dell PC)	↑	Access	VLAN 10	-
5 (MacBook Air)	↑	Access	VLAN 20	-
6 (ASR1001-X)	↑	Access	VLAN 20	-

(All links up and operating correctly)



Useful Commands

```
# Ping test
ping 192.168.20.10

# Traceroute test
tracert 8.8.8.8

# Check ARP table
arp -a

# Check FortiGate policy hit count
get firewall policy

# Check FortiGate system status
diagnose sys top
```



Key Lessons Learned

- Always verify physical connections and link status first.
- Correct VLAN assignments on switch ports are critical.
- Firewall policies must be created in the correct order.
- QUIC can bypass web filters — block UDP/443 if needed.
- Logs are essential for identifying why a policy is or isn't working.
- Document every issue and resolution for future reference.



Best Practices

- Test each change after applying configuration.
- Save configurations on all devices regularly.
- Keep a backup before making major changes.
- Use meaningful names and descriptions for policies and interfaces.
- Monitor logs and performance periodically.
- Keep documentation up to date.

11. Results & Achievements



The lab was successfully designed, implemented, tested, and documented. All objectives were achieved and validated. This project demonstrates a strong foundation in network design, security, and automation.



Project Success Summary

- ✓ Network topology designed and fully operational.
- ✓ VLANs created and segmented successfully.
- ✓ Inter-VLAN traffic controlled using firewall policies.
- ✓ Secure management access (SSH) implemented to ASR1001-X.
- ✓ Web filtering and logging configured and working.
- ✓ Python automation script created to backup configurations and gather system info.
- ✓ All tests passed and verified.



Objectives vs. Achievements

Objective	Description	Achievement	Status
Design a Segmented Network	Create a network with separate VLANs for users and lab/management.	VLAN 10 (users) and VLAN 20 (lab/mgmt) created and working.	✓ Completed
Configure Firewall Policies	Control traffic between VLANs and to the internet.	Policies created to allow/deny traffic as per requirements. Web filter applied.	✓ Completed
Secure Device Management	Enable secure SSH access to Cisco ASR1001-X.	SSH v2 enabled and working from VLAN 10 to ASR1001-X.	✓ Completed
Internet Access for User VLAN	Allow VLAN 10 to access the internet using FortiGate.	NAT and policy configured. Internet access verified.	✓ Completed
Logging & Monitoring	Enable logging for security events and web filtering.	Logs are generated and reviewed. Facebook access is blocked.	✓ Completed
Python Automation	Automate configuration backup and retrieve system information.	Python script created and tested successfully.	✓ Completed
Documentation	Document the project with diagrams, configs, tests, and troubleshooting.	Complete documentation created for GitHub portfolio.	✓ Completed



Key Results

Network Connectivity	- All devices connected and reachable. - VLANs isolated and working.	✓ Success
Inter-VLAN Control	- VLAN10 → VLAN20: Allowed - VLAN20 → VLAN10: Blocked	✓ Success
Internet Access	VLAN10 users can access the internet through FortiGate (NAT).	✓ Success
Device Management	SSH access to ASR1001-X allowed only from VLAN10.	✓ Success
Web Filtering	Facebook is blocked for both VLANs.	✓ Success
Logging	All allowed/blocked events are logged and reviewed.	✓ Success
Automation	Python script collects system info and backs up configurations.	✓ Success



What We Accomplished

- Build a secure and segmented network using VLANs.
- 1 Implemented firewall policies to control traffic between VLANs and to the internet.
 - 2 Enabled secure SSH management to the Cisco ASR1001-X.
 - 4 Configured web filtering and verified it through logs.
 - 5 Wrote a Python automation script to backup configs and gather system information.
 - 6 Documented every step with diagrams, screenshots, and explanations.
 - 7 Tested and verified all requirements successfully.



Project Impact

- Improved network security by isolating networks.
- Reduced unnecessary traffic with proper VLAN design.
- Enforced security policies to protect users.
- Gained hands-on experience with FortiGate, Cisco ASR, and managed switches.
- Built automation skills using Python and REST API.
- Created a strong portfolio project for GitHub.



Future Enhancements

- Implement Inter-VLAN Routing for communication between VLANs.
- Configure DHCP server for VLAN 10 and VLAN 20.
- Implement Site-to-Site VPN to a remote network.
- Monitor network with SNMP and Syslog server.
- Extend automation script to push configurations and create reports.
- Add OSPF routing with Cisco router for advanced scenarios.

